# MANGU 2025 KCSE PREDICTION

## CONFIDENTIAL EXAM

**451/1**

# COMPUTER STUDIES

**PAPER 1 (THEORY)**

**TIME: 2½ HOURS**

**NAME**……………………………………………………………………………………

**SCHOOL**…………………………………………………… **SIGN**………………………

**INDEX NO**…………………………………………………… **ADM NO**……………………

*(Kenya Certificate of Secondary Education)*

**INSTRUCTIONS TO CANDIDATES**

*(a) Write your Name and Index/Adm Number in the spaces provided.*
*(b) This paper consists of TWO sections: A and B*
*(c) Answer ALL the questions in section A*
*(d) Answer question 16 and any other* ***THREE*** *questions from section B*
*(e) All answers should be written in the spaces provided in the question paper.*

**For Examiners use only.**

| Section | Question | Candidates Score |
|---|---|---|
| A | 1 – 15 | |
| B | 16 | |
| | 17 | |
| | 18 | |
| | 19 | |
| | 20 | |
| Total Score | | |

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## SECTION A (40 MARKS)

## ANSWER ALL QUESTIONS FROM THIS SECTION

**1.** Give **FOUR** characteristics of first generation computers? **(2marks)**

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**2. a**) Why are powder fire extinguishers not allowed in the computer room? **(1 mark)**

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b) What is the difference between delete and backspace keys? **(2 marks)**

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**3.** Give **FOUR** factors that describe the ideal environment for a computer to work properly. **(2marks)**

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**4.** Give **TWO** advantages of using sound output devices **(2marks)**

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**5**. Give **TWO** application areas of OMR **(2marks)**

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**6. a)** Distinguish between primary and secondary storage devices. **(2marks)**

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**7.** State FOUR devices that support multimedia capability on a computer **(2marks)**

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**b)** Differentiate between the terms: volatile memory and non-volatile memory. **(2marks)**

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**8.** Differentiate between misreading and transposition errors in data processing **(2marks)**

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**9.** What is the function of the power supply unit found in the system unit? **(2marks)**

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**10.** **(a)** Name **FOUR** approaches that may be used to replace a Computerized Information system. **(2 marks)**

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**(b)** Which of the approaches named in (a) above is appropriate for critical systems? Explain. **(2 marks)**

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**11.** Terry intends to download a movie from the Internet. State the **two** factors that may determine the total time taken to complete the download. **(2 Marks)**

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**12.** A lecturer keeps the following student details in a database: name, age, course.

(a) Write an expression you would use to compute the year of birth of a student using this year as the current year. **(2 marks)**

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(b) What query expressions would the lecturer use to list the students whose age is above 15 years and below 25 years? **(2 marks)**

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**13. a)** Explain each of the following roles performed by an operating system

**i.** Memory management **(2 marks)**

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**ii.** Interrupt handling **( 2 marks)**

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**14.** A computer teacher has put a rule that flash disk should not be used in the computer laboratory

(a) Give a reason for the rule **(1 mark)**

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**(b)** State two alternatives that can be used to achieve the same objective **(1 marks)**

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**15. (a)** What is a utility software? **(1 mark)**

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**(b)** Give **FOUR** examples of utility software **(2 marks)**

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## SECTION B (60 MKS)

**Answer question 16 and any other 3 questions.**

**16. a) i)** What is system maintenance? **(2marks)**

**b)** Differentiate between absolute and relative cell referencing. **(2marks)**

**c) The** formula $A$4 +C$5 is typed in cell D4 and copied to cell H6. Write down the formula as it would appear in cell H6. **(2marks)**

d) Generate output for the following pseudo code **(4marks)**

```
START
    X= 5
    WHILE (X<15) DO
        Y= X +2
X=X+3
PRINT X,Y
    ENDWHILE
    PRINT ("DONE',X)
STOP
```

e) Draw a flowchart for the above pseudo code **(5 marks)**

**17. a)** Using 10-bits binary number system, perform the following decimal operation using two's complement. **(4 marks)**

$129_{10} - 128_{10}$

b) Convert the following into binary $6CD_{16}$ **(2 marks)**

c) Perform the following binary arithmetic 11001.101+1110.011 **(2 marks)**

d) Define the following terms. **(3 marks)**

i) Record

ii) File

iii) Database

e) List any **FOUR** ways of dealing with a virus on a computer. **(4 marks)**

**18. a)** Explain the functions performed by **(2 marks)**

**i)** The control unit-

**ii)** Arithmetic and logic unit (ALU)

b) Define the following terms in relation to computer software . (3 marks)

i) Free ware.

ii) Proprietary

iii) Open source

c) i) Define the term E-commerce **(2marks)**

ii) List down **TWO** advantages of e- commerce as used in modern business environment **(2marks)**

d) Describe **THREE** ways how computing has been applied to each of the following areas:

i) Reservation systems. **(3 marks)**

ii) Law enforcement **(3 marks)**

**19.** **(a) i**) Name any **TWO** types of non-printable guides in DTP. **(1mark)**

ii) Explain the functions of each of non-printable guides in (19 (a)i) above **(4 marks)**

b) State FOUR factors to consider when designing a good file **(2 marks)**

c) Name **THREE** responsibilities that are carried out by a:

(i) Web administrator **( 3 marks)**

(ii) Computer trainer **(3 marks)**

d) Identify **TWO** disadvantages of Observation method used in fact-finding. **(2 marks)**

**20.(a)** List **THREE** advantages of wireless communication over wired communication. (**3 marks)**

(b) Using a Diagram, describe the following signals and state where each is applied in network communication:

(i) Analog; **(2 marks)**

(ii) Digital. **(2 marks)**

(c) Name the **TWO** types of coaxial cables. **(2 marks)**

(d) (i) Define the term network protocol. **(1 mark)**

(e) List **FOUR** internet protocols **(2 marks)**

(f) Describe star topology as used in networking **(3 marks)**

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